

IBM Applied Data Science Capstone Project Report

SpaceX Falcon 9 First Stage Landing Prediction

TEMPLATE — replace all TODO sections with your actual IBM lab outputs before submission.

1. Executive Summary

TODO: Summarize methods and actual key results.

2. Introduction

TODO: Explain background, problem, and objective.

3. Data Collection – SpaceX API

TODO: Add API flowchart/screenshots and GitHub URL.

4. Data Collection – Web Scraping

TODO: Add scraping process/screenshots and GitHub URL.

5. Data Wrangling Methodology

TODO: Explain cleaning and processing.

6. EDA with Data Visualization

TODO: Add required charts and findings.

7. EDA with SQL

TODO: Add required SQL queries/results.

8. Interactive Visual Analytics – Folium

TODO: Add map, launch markers, and proximity analysis.

9. Interactive Visual Analytics – Plotly Dash

TODO: Add dashboard screenshots.

10. Required EDA Visualization Results

TODO: Add scatter plots, bar charts, and yearly trends.

11. Required SQL Results

TODO: Add launch sites, payloads, success rates, rankings, and time analysis.

12. Predictive Analysis

TODO: Add multiple models, evaluation, confusion matrix, model explanation, and conclusion.

13. Conclusion

TODO: Add evidence-based conclusions from your actual results.

14. GitHub Repository

TODO: Add your actual GitHub repository URL.